# TRIVIALITY OF AN INVERTIBLE SHEAF ON A FAMILY PRODUCT FROM TRIVIALITY ALONG THE FACTORS AND THE SPECIAL FIBRE

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 9621-9632 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9634-9650, the complete author proof (15 body lines).

*Editorial note.* Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The Kunneth injectivity for H^1 of O on a product of proper schemes over a field (Varieties), and the chapter's relative-Picard lemma promoting fibrewise triviality plus an injectivity condition on H^1(O) to local triviality; cited inputs with the composition written in the retained span.

*Source reference.* Original lemma 0BF2: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9519–9541. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BDP: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9199–9215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BED: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9650–9650. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : X \to S$ and $g : Y \to S$ be morphisms of schemes satisfying the hypotheses of Lemma 0BDP. Let $\sigma : S \to X$ and $\tau : S \to Y$ be sections of $f$ and $g$. Let $s \in S$. Let $\mathcal{L}$ be an invertible sheaf on $X \times_S Y$. If $(1 \times \tau)^*\mathcal{L}$ on $X$, $(\sigma \times 1)^*\mathcal{L}$ on $Y$, and $\mathcal{L}|_{(X \times_S Y)_s}$ are trivial, then there is an open neighbourhood $U$ of $s$ such that $\mathcal{L}$ is trivial over $(X \times_S Y)_U$.*

*Proof.* By Künneth (Varieties, Lemma 0BED) the map

$$H^1(X_s \times_{\mathrm{Spec}(\kappa(s))} Y_s, \mathcal{O}) \to H^1(X_s, \mathcal{O}) \oplus H^1(Y_s, \mathcal{O})$$

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is injective. Thus we may apply Lemma 0BF2 to the two morphisms

$$1 \times \tau : X \to X \times_S Y \quad \text{and} \quad \sigma \times 1 : Y \to X \times_S Y$$

to conclude. □